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The fragility of deterrence in conflicts

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Abstract

We study deterrence in sequential move conflicts, modeled as a contest. We bias the model in favor of peace by assuming that under complete information deterrence is achieved and peace prevails. We show that under incomplete information about states' types (resolve) the chances of deterrence decrease rapidly. Studying a uniform type distribution, we show that the finer the type space becomes the more resolve a defending state must have to support deterrence in equilibrium. In the limit, as types occur on a continuum, deterrence is possible only under relatively extreme conditions.

Keywords

Conflict; contest; deterrence; information

1. Introduction

In Stanley Kubrick's famous 1964 film *Dr Strangelove* the world is close to annihilation after a confused air-force general ordered a nuclear first strike against the Soviet Union. The Soviets, in an effort to prevent exactly such a scenario, had developed a doomsday machine, a device that consisted of 50 nuclear bombs that could not be un-triggered and were set to detonate should a nuclear attack hit the country. This machine was a commitment to mutually assured destruction and was planned to function as a deterrent.

> **President Merkin Muffley:** *How is it possible for this thing to be triggered automatically and at the same time impossible to untrigger?*

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> **Dr Strangelove:** *Mr. President, it is not only possible, it is essential. That is the whole idea of this machine, you know. Deterrence is the art of producing in the mind of the enemy the fear to attack. And so, because of the automated and irrevocable decision-making process which rules out human meddling, the Doomsday machine is terrifying and simple to understand and completely credible and convincing.*

Knowing that a first strike would directly fire back and mutual destruction was assured, a rational political leader would not give such devastating orders in the first place.

Scholarship has often attributed open conflict to the failure of deterrence: see for example Grossman and Kim (1995), Zagare and Kilgour (2000), or the recent survey of deterrence theories by Quackenbush (2011). The causes for such a failure are often believed to be commitment problems or informational asymmetries. The mechanism we explore in this paper is related to both. However, although being quite intuitive and simple, this mechanism has not yet been studied. The mechanism is demonstrated most easily by way of an example. Assume there are two states, 1 and 2, which potentially may engage in violent conflict. In particular, assume state 2 has an interest in attacking state 1 to secure itself a given rent. State 1 can now invest in its armed forces to deter state 2. Increasing armed forces makes it less likely that 2 will succeed, makes success more costly to succeed because of a prolonged war, or gives 1 the opportunity to retaliate. In any case, 1 makes it more costly for 2 to start a conflict. Whatever the relative strength of the two states or the relative resolve, if 1 invests sufficiently in armed forces it can deter 2, because engaging in conflict becomes too costly and hence not worthwhile. Assume under complete information deterrence happens and the outcome is peace, independent of 2's resolve. Of course, the assumption of complete information is quite strong, especially because the resolve of a state to engage in international conflict depends crucially on domestic politics, which change frequently over time. It is hence much more likely that 1, when choosing the optimal size of its armed forces, has a distribution of 2's resolve in mind. Now given that 1 prefers to deter 2 regardless of resolve under full information, surely 1 will also deter 2 when he is uncertain of his opponents' resolve. The latter intuition turns out to be fundamentally flawed. We show that in the presence of uncertainty 1 typically chooses to deter only a share of possible opponents, while leaving the defenses low enough to provoke an attack by possible high-resolve opponents.

What is the intuition for this finding? Consider an investment by state 1 just sufficient to deter a state 2 with the highest possible resolve. Under full information, a marginal decrease of this investment immediately assures an attack by state 2. Under uncertainty on the other hand, an attack is still highly unlikely. In all situations where state 2 does not exhibit maximal resolve deterrence still works. This means that an investment to deter even a possible maximum-resolve opponent becomes relatively costly compared to its, now uncertain, benefits. Thus state 1 optimally chooses lower investment relative to full information whenever the probability of facing the highest-resolve opponent is sufficiently small. More generally we show that the optimality of full deterrence crucially depends on the precision of information available to state 1. As the number of possible resolve 'types' becomes higher, or as information becomes less precise, the probability of full deterrence declines.

Note that the reason for possible conflict is then that states are willing to gamble or to bet on a weak opponent. Hence, we may describe such a behavior as *defensive*

brinkmanship. Contrary to 'normal' brinkmanship, which means a state engages in very risky offensive tactics of pushing an opponent state to the brink in order to force it to make concessions, defensive brinkmanship, as we find it here, relates to taking the risk of being attacked in light of an analysis of expected costs and benefits of deterrence.

The rest of the paper is organized as follows: next we discuss related literature. In Section 2 we describe the model and develop the main arguments and results. A short discussion of our model follows in Section 3. Section 4 concludes. All proofs are relegated to the appendices.

1.1. Related literature

There is an extensive body of theoretical literature by now that studies conditions for successful deterrence in international relations, recently surveyed by Quackenbush (2011). The bulk of this literature is concerned with 'rationalist' explanations (Fearon, 1995), where the possible occurrence of conflict is ascribed to rational decision making of the parties in situations where war is costly. Zagare and Kilgour (2000) further differentiate between classical deterrence theory and what they introduce as 'perfect deterrence theory,' which in the former rationality is understood as an instrumental construct while in the latter it is used as a procedural decision rule.[1] We apply the procedural rationality construct, and thus our analysis best pertains to the realm of the latter class of models.

Zagare and Kilgour (2000) also distinguish between mutual and unilateral deterrence. Our model can be interpreted as a formalization of unilateral deterrence, i.e. a situation in which only one state wants to challenge the status quo, which the other state wants to defend. If we extend the model to make the distinction between offensive and defensive weapons, as for example Grossman and Kim (1995), our result also carries over to mutual deterrence.

Deterrence models typically involve choices between the two alternatives 'conflict' and 'peace' (Quackenbush, 2011). Two strands of literature extending this binary framework have emerged: bargaining models (e.g. Powell (1996) and Slantchev (2003)) and guns-vs-butter models (e.g. Powell (1993) and Skaperdas (1992)). In the former strand, parties bargain over the revision of the international status quo with the option of launching an attack and possibly triggering war is that bargaining fails. Of great interest are conditions under which no state chooses to launch such an attack. In the latter strand, parties face a trade-off between using their resources for consumption, which directly benefits the population or elites, and investing in military armament. This armament yields no direct (consumption) utility but helps shield resources from attack or acquires resources for consumption from the rival. Our model belongs to this class of models. Like Powell (1993) and Skaperdas (1992), we are interested in finding conditions under which states refrain from attack and peace prevails. However, in contrast to these authors our focus is on the importance of asymmetric information and we show how this inhibits the incentives to deter a rival and thus increases the likelihood of conflict.

While asymmetric information has frequently been put forward as the reason for conflict, it usually is then that potential opponents are overly optimistic. Fearon (1995) and Slantchev (2010) are important exponents of conflict justifications in this fashion. Incomplete information leads to strategic behavior to manipulate the opponent's beliefs and hence can bring about conflict. Similarly, incomplete information can prevent peace

because it creates incentives to preemptive action as in Chassang and i Miquel (2010). We consider a different channel. In our paper incomplete information depresses incentives to commit resources, which leads to imperfect deterrence. Thus, contrary to Chassang and i Miquel (2010), results are driven by reduced incentives to dampen predatory motives of an opponent state.

Fearon (1995) distinguishes two explanations for conflict related to informational problems, disagreement about relative power and uncertainty about an opponent's willingness to fight. In our modeling set-up this distinction is not very meaningful, because a state's willingness to fight directly impacts relative power via military spending. However, the mechanism leading to a higher chance of conflict that we identify is quite similar in flavor to the risk–return trade-off Fearon (1995) discusses when there is uncertainty about an opponent's willingness to fight. When two states bargain over how to divide a given rent, there will usually be agreement if there is complete information and the consequence of bargaining failure is war. Because states do not know how much they can demand, this changes under asymmetric information and war becomes more likely. As Fearon puts it: 'The larger the grab, the greater the risk of war, but the better off [state] A will be if state B acquiesces' (1995: 394). Powell (1996) shows that such a trade-off also exists when states have private information about their costs of going to war. In this paper we show that there is also a risk–return trade-off in military spending. While the underlying mechanism driving this result is quite different, the result is similar and we point out another important link between deterrence and uncertainty that should be of intuitive appeal to scholars in this field.

In his classic book, Schelling (1960) identifies the credible threat to inflict pain upon the opponent in case of an attack as a means of deterrence. Other authors have gone further with this idea and explored conditions under which committing to military capacity can be sufficient to establish a peaceful resolution of conflict, for example Grossman and Kim (1995), Kolmar (2008), or Hoffmann (2010). They find that being relatively strong or having a relatively high resolve can enable a defending state to achieve deterrence, because it decreases the attacking states' chances of success. We show that under incomplete information a state often does not have the incentive to commit sufficient resources so that with some positive probability the pain inflicted on the attacker is not enough for deterrence. Thus previous results become much more nuanced and have to be interpreted much more carefully in light of informational asymmetries.

In a related paper, McBride et al. (2011) show how sufficient state capacity can foster peace, because this decreases the rents appropriable by an intruder or attacking state. Our analysis suggests that states often do not have incentives to invest in sufficient state capacity if there is incomplete information.

Powell (2007) studies how a state commits resources in a conflict under incomplete information. However, in his paper the defending state knows more and has to decide how to allocate armies to different battlefields in a Colonel Blotto game, when sending armies might reveal the vulnerability of a battlefield. In our paper the informational advantage is with the potential attacker and, unlike Powell, we explore how this can impede deterrence.

Also starting with Schelling (1960), scholars argued that commitment may increase the likelihood of overt conflict because it may lead to bargaining failure. See also the works by Crawford (1982) or more recently Ellingsen and Miettinen (2008). Contrary to these papers, in our paper commitment does not relate to bargaining strategy, but to

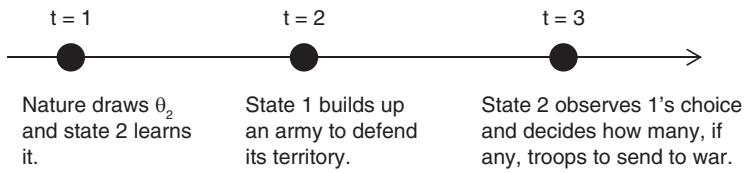

Figure 1. Sequence of moves.

resources that are devoted to increasing a state's military strength and hence the ability to deter an opponent.

Finally, we contribute to the literature analyzing contest and conflicts, surveyed by Konrad (2009) and Garfinkel and Skaperdas (2007). The implications of informational asymmetries in contest games have been the subject of study of many authors. The emphasis is usually on simultaneous move games, e.g. Wärneryd (2003), Fey (2008), or Hurley and Shogren (1998a,b). We, in contrast, study a sequential move game. Sequential move games with asymmetric information have been studied by Linster (1993) and Fu (2006). Linster (1993) characterizes the equilibrium of a sequential rent-seeking contest while Fu (2006) studies how players endogenously determine the order of moves under incomplete information and shows that the better informed agent will move second. Our contribution is to study the effectiveness of deterrence in sequential move games when states are asymmetrically informed using a generalization of the model by Linster (1993).

2. Model

Two states $i = 1, 2$ compete for a fixed, indivisible rent. Hence, bargaining is not possible and the conflict has to be resolved through fighting. To increase the chance of success in the fight, states invest in guns $x_i \geq 0$. Buying guns is costly, representing the usual guns-vs-butter trade-off (e.g. Skaperdas (1992) or Powell (1993)). State 1 is able to commit to the level of guns while 2 is not. This implies that state 2 will react to state 1's investment when making its choice. This may for example represent a first mover advantage of 1. Guns are observable. For example, building a wall around a city or house and moving troops are usually observable. At the time of this first move, state 1 does not know all relevant information about state 2. For example, it does not know 2's military capacity, how advanced its weaponry is, or how it values winning the conflict. To the contrary, after 1's move 2 observes all relevant information. Hence, there are informational asymmetries. To model this asymmetry we define θ_i as state i's type. It represents the (idiosyncratic) value a state attaches to winning the conflict and hence relates to a state's resolve. θ_1 is commonly known at the time of 2's decision, but when 1 decides about its action it only has a prior about θ_2. In particular, it believes its opponent's type to be drawn from $[\underline{\theta}_2, \overline{\theta}_2] \subset \mathbb{R}_+$ with density $p_2(\theta_2)$. Without loss of generality, the type of a state relates to the value it assigns to winning the conflict, its resolve. Figure 1 shows the sequence of moves. In Section 3 we shortly discuss alternative timing assumptions.

Both states can invest in the conflict by building up a stock of guns or spending effort, x_1 and x_2. The larger the stock of guns x_i of a state, the higher its chances of success in the

conflict. To fix ideas we assume the following relationship between guns and the states' probability of winning the conflict:

Assumption 2.1. *State 1 wins with probability*

$$q_1(x_1, x_2) = \frac{f_1(x_1)}{f_1(x_1) + f_2(x_2)} \tag{1}$$

where $f_i'(x_i) > 0$, $f_i''(x_i) \leq 0$, $f_1(0) > 0$, $\lim_{x\to\infty} f_1(x) = \infty$, *and* $f_2(0) = 0$. *2 wins with the remaining probability,* $q_2 = 1 - q_1$.

This is a standard function to determine conflict outcomes.[2] $f_1(0) > 0$ and $f_2(0) = 0$ imply that even without investing in defense, 1 has a chance of keeping possession, while 2 will need to expend a positive effort to have a chance. This reflects 1's initial possession of a good or 1 being the incumbent in a given territory. In fact, it means that if state 2 chooses $x_2 = 0$, state 1 keeps the disputed good with probability $q_1 = 1$. $\lim_{x\to\infty} f_1(x) = \infty$ guarantees that deterrence is technologically always feasible.

States are not risk loving and have standard Von Neumann–Morgenstern utility functions:

Assumption 2.2. *State i has the following utility function:*

$$U_i(x_i, x_j; \theta_i) = q_i(x_i, x_j)u_i(\theta_i - x_i) + (1 - q_i(x_i, x_j))u_i(-x_i) \tag{2}$$

where $u_i'(.) > 0$ *and* $u_i''(.) \leq 0$. *Furthermore we assume that* $q_i(.)$ *and* $u_i(.)$ *are such that* $U_i''(.) < 0$.[3]

Since state 1 does not know its opponent's type it also does not know its equilibrium behavior. Hence, its expected utility takes the following form:

$$EU_1(x_1, x_2; \theta_1) = E_{\theta_2}[q_1(x_1, \bar{x}_2(x_1; \theta_2))]u_1(\theta_1 - x_1) + (1 - E_{\theta_2}[q_1(x_1, \bar{x}_2(x_1; \theta_2))])u_1(-x_1) \tag{3}$$

where $\bar{x}_2(x_1; \theta_2)$ is 2's best response to x_1 if its type is θ_2. Therefore we now turn to this best response. Upon observing 1's effort choice, 2 maximizes

$$U_2(x_1, x_2; \theta_2) = q_2(x_1, x_2)u_2(\theta_2 - x_2) + (1 - q_2(x_1, x_2))u_2(-x_2)$$

This yields

Lemma 2.3. *State 2's best response is implicitly defined by*

$$\bar{x}_2(x_1; \theta_2) = \max\left\{0, x_2 \in (0, \theta_2) : \frac{f_1(x_1)f_2'(x_2)\Delta u_2}{f_1(x_1) + f_2(x_2)} = f_2(x_2)u_2'(\theta_2 - x_2) + f_1(x_1)u_2'(-x_2)\right\}$$

where $\Delta u_2 = u_2(\theta_2 - x_2) - u_2(-x_2)$. $\bar{x}_2(x_1; \theta_2)$ *is flat* $(= 0)$ *for low* θ_2 *and strictly increasing in* θ_2 *thereafter.*

Proof. See Appendix A. □

For deterrence of type θ_2 it needs to hold that $\bar{x}_2(x_1; \theta_2) = 0$ in equilibrium. For 1 to deter a type with higher value it needs to spend more:

Lemma 2.4. *For any $\theta_2 \in [\underline{\theta}_2, \overline{\theta}_2]$ there exists $\mu(\theta_2)$ such that if $x_1 \geq \mu(\theta_2)$ the best response of 2 is $\bar{x}_2(x_1;\theta_2) = 0$. Moreover, $\mu'(\theta_2) > 0$, implying that in order to deter a higher type higher effort is necessary.*

Proof. See Appendix B. □

We assume that state 1's resolve θ_1 is sufficiently large that under complete information in equilibrium it will choose $x_1 = \mu(\overline{\theta_2})$ against a state 2 with resolve $\overline{\theta_2}$, and hence peace always prevails. The purpose is to show that it is not the distribution of power that drives our result but uncertainty.

Now look at state 1's incentives. Maximizing utility, it takes into account 2's best response. Of course, there are many different types of 2. To simplify things, assume for the moment the type space is discrete with constant density. There are $n \geq 2$ types that occur with equal probability. In particular, the type space is the following:

Assumption 2.5. *There are $n \geq 2$ types. The type space (grid) is given by*

$$\Theta_2 = \left(\underline{\theta}_2 + \frac{k}{n-1}(\overline{\theta}_2 - \underline{\theta}_2)\right)_{k=0}^{n-1}$$

and each type occurs with probability $1/n$.

For deterrence we need the highest type to remain passive in equilibrium:

$$\bar{x}_2(x_1;\overline{\theta}_2) = 0 \Leftrightarrow x_1 \geq \mu(\overline{\theta}_2)$$

Look at state 1's problem when it decides on optimal effort, taking into account state 2's reaction $\bar{x}_2(x_1;\theta_2)$, assuming $(n-1)$ types are already deterred:

$$\max_{x_1 \in [0,\theta_1]} U_1(x_1, x_2(x_1;\theta_2);\theta_1) = \frac{n-1}{n} u_1(\theta_1 - x_1) + \frac{1}{n}\left(\frac{f_1(x_1)u_1(\theta_1 - x_1)}{f_1(x_1) + f_2(\bar{x}_2(x_1;\overline{\theta}_2))} + \frac{f_2(\bar{x}_2(x_1;\overline{\theta}_2))u_1(-x_1)}{f_1(x_1) + f_2(\bar{x}_2(x_1;\overline{\theta}_2))}\right)$$

Taking (left) derivatives the first-order condition at $x_1 = \mu(\overline{\theta}_2)$, the point of deterrence, is

$$\frac{\partial U_1(x_1, x_2(x_1;\theta_2);\theta_1)}{\partial x_1}\Big|_{x_1=\mu(\overline{\theta}_2)} = -u_1'(\theta_1 - \mu(\overline{\theta}_2)) - \frac{f_2'(0)}{nf_1(\mu(\overline{\theta}_2))}\frac{\partial \bar{x}_2(x_1;\overline{\theta}_2)}{\partial x_1}\Big|_{x_1=\mu(\overline{\theta}_2)} \Delta u_1(\theta_1 - \mu(\overline{\theta}_2)) \geq 0 \tag{4}$$

where $\frac{\partial \bar{x}_2(x_1;\overline{\theta}_2)}{\partial x_1}$ is the left derivative of $\bar{x}_2(x_1;\overline{\theta}_2)$ with respect to x_1 (see equation (6) in Appendix B). $\mu(\overline{\theta}_2)$ looks as follows:

Lemma 2.6. *For deterrence of θ_2, 1 needs to spend at least*

$$\mu(\theta_2) = f_1^{-1}\left(\frac{\Delta u_2(\theta_2, 0) f_2'(0)}{u_2'(0)}\right)$$

Proof. See Appendix C. □

The derivative is as follows:

$$\frac{\partial \bar{x}_2(x_1;\overline{\theta}_2)}{\partial x_1}|_{x_1=\mu(\overline{\theta}_2)} = \frac{\frac{f_2{}'(0)f_1{}'(\mu(\overline{\theta}_2))}{f_1(\mu(\overline{\theta}_2))^2}\Delta u_2(\overline{\theta}_2,0)}{\frac{f_2{}''(0)f_1(\mu(\overline{\theta}_2))-2f_2{}'(0)^2}{f_1(\mu(\overline{\theta}_2))^2}\Delta u_2(\overline{\theta}_2,0)+u_2{}''(0)} < 0$$

Hence, we can rearrange (4) in the following way to get a relation between the precision of information and deterrence in equilibrium:

$$n \leq -\frac{f_2{}'(0)}{f_1(\mu(\overline{\theta}_2))u_1{}'(\theta_1-\mu(\overline{\theta}_2))}\frac{\partial \bar{x}_2(x_1;\overline{\theta}_2)}{\partial x_1}|_{x_1=\mu(\overline{\theta}_2)}\Delta u_1(\theta_1-\mu(\overline{\theta}_2))$$

To bias the model towards peace means that this condition holds at least if $n = 1$. The right-hand side is strictly positive. What the condition now tells us is that the larger n becomes, the less likely it is that there is deterrence. In other words: if the condition holds with equality and we increase n, θ_1 needs to be increased as well for the condition to hold.

Put differently, if θ_1 is such that the first-order condition holds with equality at deterrence, increasing n has the effect that deterrence is not an equilibrium anymore. Proposition 2.7 formalizes the result:

Proposition 2.7. *The finer the grid of types, Θ_2, i.e. the larger n, the higher θ_1 needs to be compared to $\overline{\theta}_2$ for deterrence in equilibrium.*

Proof. Follows from the discussion above. □

To get a feeling for what that means exactly, let $f_i(x_i) = x_i$, and $u_i(x) = x$. It is then easily verified that a sufficient and necessary condition for deterrence is $\theta_1 \geq 2n\overline{\theta}_2$ in case of a uniform distribution or $\theta_1 \geq \frac{2\overline{\theta}_2}{Pr[\theta_2=\overline{\theta}_2]}$ for a general discrete distribution. Even if there are only four different uniformly distributed types of 2, 1 would need a type θ_1 that is (at least) eight times as high as the highest type of 2. Deterrence hence becomes increasingly unlikely. What does this imply for the probability of conflict? It is easy to see that it can increase significantly. For example, let states be risk neutral and the probability of success be given by $q_1(x_1,x_2) = \frac{c+x_1}{c+x_1+x_2}$ for $c \to 0$. This is then the same technology as used in Esteban and Ray (2008). Further, let $\Theta_2 = \left\{\frac{\theta_1}{4}, \frac{\theta_1}{3}, \frac{\theta_1}{2}\right\}$, which implies that under complete information deterrence of all types is optimal and peace always prevails. Under incomplete information this result breaks down. State 1's equilibrium strategy is $x_1 = \frac{1}{36}\left(5+2\sqrt{6}\right)\theta_1$, which implies that only the lowest-resolve opponent is deterred and the probability of conflict increases from zero to 2/3. State 1's probability of losing the conflict conditional on being attacked is $\frac{7}{12} - \frac{1}{\sqrt{6}} \approx .175$. Hence, defensive brinkmanship can be significant.

The above discussion also directly implies the following result:

Corollary 2.8. *Let the distribution of types have density $g(\theta_2) > 0$ with continuous support $[\underline{\theta}_2, \overline{\theta}_2] \subset \mathbb{R}_+$. There is never deterrence in equilibrium and 2 is active with a strictly positive probability. Hence, there is $\theta_2' < \overline{\theta}_2$ such that all types $\theta_2 > \theta_2'$ engage in conflict.*

Proof. See Appendix D. □

The fraction of types that are active is always strictly positive. The consequence is that under the assumptions made deterrence is unlikely when it is not clear what type the opponent is exactly. Also note that this does not depend on the width of the type space, $\overline{\theta}_2 - \underline{\theta}_2$. Even when the different types are similar to each other, our results apply.

How robust are our findings? With respect to the utility functions, we did not make any assumptions except that they are of the Von Neumann–Morgenstern type. Also the type distribution is general. Hence, we now study what happens if the contest success function $q_1(x_1, x_2)$ changes. In particular, we look at the most extreme case in which winning probabilities are not a smooth function of efforts but jump from zero to one when one state spends more than the other. In the parlance of auction theory such a technology is called an All-pay auction (see e.g. Hillman and Riley (1989) or Baye et al. (1996)). Winning probabilities are now[4]

$$q_1(x_1, x_2) = \begin{cases} 1 & \text{if } x_1 > x_2 \\ 0 & \text{if } x_1 \leq x_2 \end{cases}$$

Is it now possible to achieve deterrence in this limit case of a contest if the type space is continuous? Assuming 2 only expends effort if that yields higher utility than staying passive, the best response of 2 is

$$\bar{x}_2(x_1; \theta_2) = \begin{cases} x_1 & \text{if } x_1 < \theta_2 \\ 0 & \text{else} \end{cases}$$

Hence, given effort x_1, the chance to win is exactly $G(x_1)$, and 1's expected utility is

$$EU_1(x_1, x_2; \theta_1) = G(x_1)u_1(\theta_1 - x_1) + (1 - G(x_1))u_1(-x_1)$$

The first-order condition is[5]

$$\begin{aligned} \frac{\partial EU_1(x_1, x_2; \theta_1)}{\partial x_1} &= g(x_1)u_1(\theta_1 - x_1) - G(x_1)u_1'(\theta_1 - x_1) \\ &\quad - g(x_1)u_1(-x_1) - (1 - G(x_1))u_1'(-x_1) \geq 0. \end{aligned}$$

Deterrence, i.e. $x_1 = \mu(\overline{\theta}_2) = \overline{\theta}_2$, can be an equilibrium iff $\theta_1 \geq \overline{\theta}_2$ and

$$\begin{aligned} & g(\overline{\theta}_2)\left[u_1(\theta_1 - \overline{\theta}_2) - u_1(-\overline{\theta}_2)\right] - u_1'(\theta_1 - \overline{\theta}_2) \geq 0 \\ & \Leftrightarrow g(\overline{\theta}_2) \geq \frac{u_1'(\theta_1 - \overline{\theta}_2)}{u_1(\theta_1 - \overline{\theta}_2) - u_1(-\overline{\theta}_2)} > 0 \end{aligned} \tag{5}$$

If g is differentiable on $[\underline{\theta}_2, \overline{\theta_2}]$, it must be zero at the boundaries, and hence deterrence cannot be an equilibrium. In this case the result in Corollary 2.8 still holds: there is no deterrence and a strictly positive fraction of types will invest in the conflict. However, if this is not the case, the above inequality (5) shows a condition under which deterrence can actually occur. If, for example, θ_2 is distributed uniformly on $[\underline{\theta}_2, \overline{\theta}_2]$ the density is $1/(\overline{\theta}_2 - \underline{\theta}_2) > 0$ at $\overline{\theta}_2$. Using this in the condition yields

$$\frac{u_1(\theta_1 - \overline{\theta}_2) - u_1(-\overline{\theta}_2)}{u_1'(\theta_1 - \overline{\theta}_2)} \geq \overline{\theta}_2 - \underline{\theta}_2$$

Thus, when $\overline{\theta}_2 \to \underline{\theta}_2$ and uncertainty vanishes, there will be deterrence iff $\theta_1 > \overline{\theta}_2$. We summarize the results in the following proposition:

Proposition 2.9. *Deterrence in the All-pay auction is possible iff* $\theta_1 \geq \overline{\theta}_2$ *and at the same time the density of the highest type is sufficiently high.*

Proof. Follows from the discussion above. □

Hence, even if the contest is perfectly discriminating, deterrence is unlikely or at least demands a very particular conflict scenario.

3. Discussion

Given our finding that uncertainty may drive conflict, an immediate question is why this uncertainty cannot be overcome. In many situations sharing information is mutually beneficial. For example, in a recent paper studying a conflict between two parties, Denter et al. (2011) show that quite often the two rivals agree on whether to share private information on their resolve. Similarly, Katsenos (2009) studies pre-conflict signaling and provides conditions such that information revelation happens in equilibrium. In our setting, however, there is never revelation of information through signaling. The reason is that the attacking state clearly benefits from uncertainty. To see this remember that under complete information there is deterrence of all types. This is so because state 1's resolve is high enough to deter any potential type of state 2 by assumption. Under uncertainty we have shown that this is not optimal anymore for state 1. Thus, while some types of state 2 are still deterred from attacking, some are not, and for these the presence of informational asymmetries is beneficial. After all, by opting to attack while having the option to abstain from war, we can infer that state 2 is weakly better off in the presence of uncertainty. Thus, the rivals will never agree to share information about state 2's resolve.

It is also worth noting at this point that without the opportunity to commit resources, in this class of models (guns-vs-butter) there cannot be deterrence in a Nash equilibrium. To see this, assume states move simultaneously instead. Deterrence means that the potentially attacking state sends zero troops. Without the opportunity to commit resources to defense, the best reaction for the defending state in such a situation would be not to have troops as well, because troops are costly and without having to fear an attack there is no need for their own army. Hence, often there cannot be deterrence without commitment in guns-vs-butter models (see also the discussion in Grossman (2001)). However, extending the model to defensive as well as offensive investment/weapons, where both states first choose defensive investments simultaneously, then, after observing each others' choices, decide on their offensive effort, makes deterrence—both unilateral and mutual as discussed in Zagare and Kilgour (2000)—possible in equilibrium. In particular, deterrence can come only from defensive guns for a reason similar to the one described in the above paragraph and our results carry over. In such a situation, when the resolve to attack is a state's private information, the exact logic we described above will again make deterrence less likely.[6]

4. Conclusion

We developed a simple model of conflict to illustrate how asymmetric information about the resolve of an opponent impairs the incentives to invest in deterrence. We show that a state that chooses to perfectly deter another state from attacking regardless of its resolve, prefers to save on its investment under the slightest bit of uncertainty, often leaving itself vulnerable to attack by high-resolve opponents. A peaceful outcome through deterrence becomes highly unlikely.

Many generalizations are possible, but the results will qualitatively remain unchanged. For example, states usually do not only need to decide about the size of their army, but also how to allocate forces to different potential battlefields. In such a Colonel Blotto-style game our results will prevail completely. Another example is fighting in an alliance, which would probably weakly increase the set of deterred opponents, but the main result would nevertheless prevail also in this case. In an uncertain environment economic incentives are likely to impede deterrence.

Appendix A. Proof of Lemma 2.3

Proof. Before getting started we need the following preliminaries:

$$
\begin{aligned}
q_2(x_1,x_2) &= \frac{f_2(x_2)}{f_1(x_1)+f_2(x_2)} \\
\frac{\partial q_2(x_1,x_2)}{\partial x_2} &= \frac{f_2'(x_2)f_1(x_1)}{(f_1(x_1)+f_2(x_2))^2} \\
\frac{\partial q_2(x_1,x_2)}{\partial x_1} &= -\frac{f_2(x_2)f_1'(x_1)}{(f_1(x_1)+f_2(x_2))^2} \\
\frac{\partial^2 q_2(x_1,x_2)}{(\partial x_2)^2} &= \frac{f_1(x_1)\left[f_2''(x_2)(f_1(x_1)+f_2(x_2))-2(f_2'(x_2))^2\right]}{(f_1(x_1)+f_2(x_2))^3} \\
\frac{\partial^2 q_2(x_1,x_2)}{\partial x_1 \partial x_2} &= \frac{f_2'(x_2)f_1'(x_1)\left[f_2(x_2)-f_1(x_1)\right]}{(f_1(x_1)+f_2(x_2))^3}
\end{aligned}
$$

Note that the first derivative is strictly positive and that the second derivative is strictly negative as long as $f_2''(x_2)$ is sufficiently small. If it is negative it generally holds.

In Assumption 2.2 we assume that both states' utility functions are strictly concave in their own strategy x_i, $i = 1, 2$. Hence, we can use first-order conditions to determine the best response of state 2. Then, given x_1, the best response is either $x_2 > 0$ that leads to $\frac{\partial U_2(x_1,x_2;\theta_2)}{\partial x_2} = 0$ or $x_2 = 0$. Using $q_2(x_1,x_2)$ and $\frac{\partial q_2(x_1,x_2)}{\partial x_2}$ in $\frac{\partial U_2(x_1,x_2;\theta_2)}{\partial x_2} = 0$, rearranging and dividing both sides by $f_1(x_1)+f_2(x_2) > 0$ yields the expression from Lemma 2.3.

Now look at the second part of the lemma. To see that the best response is strictly increasing in θ_2 when $x_2(x_1) > 0$ we implicitly differentiate the first-order condition:

$$
\frac{\partial \bar{x}_2(x_1)}{\partial \theta_2} = -\frac{\frac{\partial q_2(\cdot)}{\partial x_2} u_2'(\theta_2 - x_2) - q_2(\cdot)u''(\theta_2 - x_2)}{\frac{\partial^2 q_2(\cdot)}{(\partial x_2)^2}\Delta u_2(\theta_2,x_2) + q_2(\cdot)u_2''(\theta_2 - x_2) + (1-q_2(\cdot))u_2''(-x_2)} > 0
$$

The denominator is exactly the second-order condition and hence negative. The numerator is strictly positive. Hence, the whole derivative must be positive. To see that for too

small θ_2 the first-order condition is negative and hence the best response is $x_2(x_1) = 0$, let $\theta_2 \to 0$. Then the first-order condition boils down to

$$-u_2{}'(-x_2) < 0$$

Hence, for small θ_2 close to zero the best response must be flat as θ_2 increases. Only once it hits zero is the best response strictly increasing. □

Appendix B. Proof of Lemma 2.4

Proof. First, note that for $f_2{}'(0)$ too low, it might happen that even without state 1 expending positive effort the best response of 2 is always $x_2 = 0$. Since this case is not particularly interesting, we assume $\bar{x}_2(0; \underline{\theta}_2) > 0$.

To prove the lemma we implicitly differentiate 2's first-order condition again:

$$\frac{\partial \bar{x}_2(x_1;\theta_2)}{\partial x_1} = -\frac{\frac{\partial^2 q_2(\cdot)}{\partial x_1 \partial x_2}\Delta u_2(\theta_2, x_2) - \frac{\partial q_2(\cdot)}{\partial x_1}\left(u_2{}'(\theta_2 - x_2) - u_2{}'(-x_2)\right)}{\frac{\partial^2 q_2(\cdot)}{(\partial x_2)^2}\Delta u_2(\theta_2, x_2) + q_2(\cdot)u_2{}''(\theta_2 - x_2) + (1 - q_2(\cdot))u_2{}''(-x_2)} \tag{6}$$

The denominator is the same as above and hence negative. Thus, the sign of the derivative is the same as the sign of the numerator. Note that $\frac{\partial q_2(\cdot)}{\partial x_1}\left(u_2{}'(\theta_2 - x_2) - u_2{}'(-x_2)\right) > 0$ and so is $\Delta u_2(\theta_2, x_2)$. Hence, if (but not only if) $f_1(x_1) > f_2(x_2)$, the whole derivative is negative. Also note that as $x_1 \to +\infty$, $\frac{\partial U_2(x_1, x_2; \theta_2)}{\partial x_2} = -u_2(-x_2) < 0$ independent of θ_2 (which was assumed to be finite). As a consequence, the best response is $\bar{x}_2(x_1; \theta_2) = 0$, which proves the lemma. □

Appendix C. Proof of Lemma 2.6

Proof. By definition, $\mu(\theta)$ is the minimum effort x_1 that induces $\bar{x}_2(x_1, \theta_2) = 0$. From the other lemmas and the first-order condition of 2, it follows that this is given by

$$\mu(\theta_2) = \left\{x_1 \geq 0 : \frac{\partial q_2(\cdot)}{\partial x_2}\Delta u_2(0) - q_2(\cdot)u_2{}'(\theta_2) - (1 - q_2(\cdot))u_2{}'(0) = 0\right\}$$

Using the specific functions we find

$$\frac{f_2'(0)}{f_1(\mu(\theta_2))}\Delta u_2(\theta_2, 0) - u_2{}'(0) = 0$$

Simple manipulations reveal that

$$\mu(\theta_2) = f_1^{-1}\left(\frac{\Delta u_2(\theta_2, 0) f_2'(0)}{u_2{}'(0)}\right)$$

□

Appendix D. Proof of Corollary 2.8

Proof. Using our results from above but with a continuous type space with distribution $G(\theta_2)$, 1's optimization problem reads as follows:

$$\left(G(\omega(x_1)) + \int_{\omega(x_1)}^{\overline{\theta}_2} q_1(x_1, \bar{x}_2(x_1;\theta_2))dG(\theta_2)\right) u_1(\theta_1 - x_1)$$
$$+ \int_{\omega(x_1)}^{\overline{\theta}_2} (1 - q_1(x_1, \bar{x}_2(x_1;\theta_2)))\, dG(\theta_2) u_1(-x_1)$$

where $\omega(x_1) \equiv \mu^{-1}(\theta_2)$ is the marginal type deterred with effort x_1. Taking the first derivative evaluated at $x_1 = \mu(\overline{\theta}_2)$, using $\omega(\mu(\overline{\theta}_2)) = \overline{\theta}_2$, $q_1(\mu(\overline{\theta}_2), \bar{x}_2(\mu(\overline{\theta}_2))) = 1$, we get

$$-u_1'(\theta_1 - \mu(\overline{\theta}_2)) < 0$$

This is the marginal cost of effort weighted by the probability to encounter a type lower than $\overline{\theta}_2$. The derivative is negative, and therefore the necessary condition for a maximum is not fulfilled. Hence, it cannot be optimal to deter type $\overline{\theta}_2$. Moreover, continuity of the derivative implies this is true not only for this type, who has measure zero, but for some interval of types $[\theta_2', \overline{\theta}_2]$ with strictly positive measure. □

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Notes

1. Formally, they employ as their solution concept Perfect Bayesian Equilibrium. See also the discussion in Zagare (2004).
2. It is a slightly more general form than in Slantchev (2010) or Esteban and Ray (2008). See Skaperdas (1996) for an axiomatization.
3. For example, if states are risk neutral this generally holds. Generally, if the stakes are high or marginal utility does not change dramatically, the condition holds.
4. We assume that 2 wins in case of a tie to guarantee existence of a best response. Alternatively, we could also assume that the smallest unit of effort is ϵ with $\epsilon \to 0$ but strictly positive.
5. This condition only characterizes the optimum if the second-order condition holds. However, if the first-order condition at deterrence does not hold, deterrence cannot be an optimum independent of the second-order condition.

6. Many scenarios are thinkable. First, in the case where both states send offensive troops, these troops will fight and determine the intermediary winner on the battlefield. This winner then gets to attack the losing state's defense. While in this situation strategic complexity is increased significantly, because attacking may serve as a second line of defense, the discussed trade-off remains important. In particular, if both states invest in sufficient defensive weaponry, no state has an incentive to attack. The incentives to invest are, however, depressed by asymmetric information as before. Second, it might be the case that offensive troops only meet the opponent's defense. In this situation the problem we studied before is just duplicated and everything remains unchanged.

References

Baye MR, Kovenock D, and de Vries CG (1996) The all-pay auction with complete information. *Economic Theory* 8: 291–305.

Chassang S and i Miquel GP (2010) Conflict and deterrence under strategic risk. *Quarterly Journal of Economics* 125: 1821–1858.

Crawford VP (1982) A theory of disagreement in bargaining. *Econometrica* 50: 607–637.

Denter P, Morgan J and Sisak D (2011) *'Where Ignorance is Bliss, 'tis Folly To Be Wise': Transparency in Contests*. Technical Report 2011-28, University of St Gallen, School of Economics and Political Science, Switzerland.

Ellingsen T and Miettinen T (2008) Commitment and conflict in bilateral bargaining. *American Economic Review* 98: 1629–1635.

Esteban J and Ray D (2008) On the salience of ethnic conflict. *American Economic Review* 98: 2185–2202.

Fearon JD (1995) Rationalist explanations for war. *International Organization* 49: 379–414.

Fey M (2008) Rent-seeking contests with incomplete information. *Public Choice* 135: 225–236.

Fu Q (2006) Endogenous timing of contest with asymmetric information. *Public Choice* 129: 1–23.

Garfinkel MR and Skaperdas S (2007) Economics of conflict: an overview. In: Hartley K and Sandler T (eds) *Handbook of Defense Economics*. Amsterdam; New York: Elsevier, Vol. 2, pp. 649–709.

Grossman HI (2001) The creation of effective property rights. *American Economic Review* 91: 347–352.

Grossman HI and Kim M (1995) Swords or plowshares? A theory of the security of claims to property. *Journal of Political Economy* 103(6): 1275–1288.

Hillman A and Riley J (1989) Politically contestable rents and transfers. *Economics and Politics* 1: 17–40.

Hoffmann M (2010) Enforcement of property rights in a barter economy. *Social Choice and Welfare* 34: 249–263.

Hurley TM and Shogren JF (1998a) Asymmetric information contests. *European Journal of Political Economy* 14: 645–665.

Hurley TM and Shogren JF (1998b) Effort levels in a Cournot Nash contest with asymmetric information. *Journal of Public Economics* 69(2): 195–210.

Katsenos G (2009) Long-term conflict: how to signal a winner? Working paper, University of Hannover.

Kolmar M (2008) Perfectly secure property rights and production inefficiencies in Tullock contests. *Southern Economic Journal* 75: 441–456.

Konrad KA (2009) *Strategy and Dynamics in Contests*. Oxford; New York: Oxford University Press.

Linster BG (1993) Stackelberg rent-seeking. *Public Choice* 77: 307–321.

McBride M, Milante G and Skaperdas S (2011) Peace and war with endogenous state capacity. *Journal of Conflict Resolution* 55: 446–468.

Powell R (1993) Guns, butter, and anarchy. *American Political Science Review* 87: 115–132.

Powell R (1996) Stability and the distribution of power. *World Politics* 48: 239–267.

Powell R (2007) Allocating defensive resources with private information about vulnerability. *American Political Science Review* 101: 799–809.

Quackenbush SL (2011) Deterrence theory: where do we stand? *Review of International Studies* 37: 741–762.

Schelling T (1960) *The Strategy of Conflict*. Cambridge, MA: Harvard University Press.

Skaperdas S (1992) Cooperation, conflict, and power in the absence of property rights. *American Economic Review* 82(4): 720–739.

Skaperdas S (1996) Contest success functions. *Economic Theory* 7(2): 283–290.

Slantchev BL (2003) The power to hurt: costly conflict with completely informed states. *American Political Science Review* 97: 123–133.

Slantchev BL (2010) Feigning weakness. *International Organization* 64: 357–388.

Wärneryd K (2003) Information in conflicts. *Journal of Economic Theory* 110: 121–136.

Zagare FC (2004) Reconciling rationality with deterrence: a reaxamination of the logical foundations of deterrence theory. *Journal of Theoretical Politics* 16: 107–141.

Zagare FC and DM Kilgour (2000) *Perfect Deterrence*. Cambridge: Cambridge University Press.